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NarrativeSense: Benchmarking Large Language Models for Socio-Political Identity Framing in Bengali News Articles

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Thesis/Project Title

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By

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Recommendation Letter from Thesis/Project Supervisor

The thesis/project entitled *“NarrativeSense: Benchmarking Large Language Models for Socio-Political Identity Framing in Bengali News”* submitted by the students

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is under my supervision. I, hereby, agree that the thesis/project can be submitted for examination.

Signature of the Supervisor: ____________

Name of the Supervisor: Mohammad Abdullah Al Mumin

Date: 23rd June, 2026

Certificate of Acceptance of the Thesis/Project

The thesis/project entitled *“NarrativeSense: Benchmarking Large Language Models for Socio-Political Identity Framing in Bengali News”* submitted by the students

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2. Md. Fuad Al Amin

on 23rd June, 2026 is, hereby, accepted as the partial fulfillment of the requirements for the award of their Bachelor Degrees.

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Abstract

The proliferation of digital news in Bangladesh has created a complex information ecosystem where media bias extends beyond simple political partisanship. In a landscape characterized by deep cultural polarization and sensitive communal dynamics, the "truth" is often distorted not by factual inaccuracy, but by strategic narrative framing.

While existing Natural Language Processing (NLP) benchmarks such as BanglaBias have made strides in detecting overt political stances, a critical technological gap remains in understanding how specific social identities are constructed, marginalized, or weaponized within discourse.

This research introduces **NarrativeSense**, a novel computational benchmark designed to detect subtle identity-based framing in Bengali news editorials. We present a curated, expert-annotated dataset and rigorously evaluate the zero-shot and few-shot reasoning capabilities of State-of-the-Art Large Language Models (e.g., GPT-4o, Gemini 1.5 Pro, Llama 3) on this task. By integrating traditional stance detection with fine-grained identity analysis, this study offers a robust pipeline for uncovering the mechanisms of polarization in low-resource languages—moving the field from detecting *what* the bias is to understanding *how* the narrative is engineered.

Keywords: Identity Framing, Bengali NLP, Large Language Models (LLMs), Media Bias, Socio-Political Analysis, Low-Resource Languages.

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Chapter 1

Introduction

News media play a central role in shaping public understanding of political events, social identities, and collective narratives. In Bangladesh, where political developments, cultural identities, and religious discourse intersect intensely—the influence of media narratives is particularly significant [1, 2]. **News outlets do not merely report events; they actively construct meaning by emphasizing certain aspects, omitting others, and portraying actors in distinct ways** [3, 4]. These processes, collectively known as framing, have profound implications for how audiences interpret social realities and form opinions.

While global research in media studies and computational linguistics has increasingly examined framing and bias, most existing work focuses on high-resource languages, especially English [5, 6]. Such studies benefit from mature NLP tools, large annotated corpora, and consistent theoretical frameworks. In contrast, Bengali (Bangla), despite being one of the most widely spoken languages in the world—remains severely underrepresented in computational media analysis [7, 8]. As a result, key dimensions of Bangladeshi news writing, including identity-based framing, remain poorly understood.

1.1 Media Landscape of Bangladesh

Bangladesh hosts a vibrant and rapidly evolving media ecosystem comprising newspapers, online portals, television networks, and digital platforms [9, 10]. The diversity of outlets reflects a wide ideological spectrum, ranging from government-aligned publications to opposition-affiliated sources. However, this plurality also introduces variations in story selection, framing strategies, and identity portrayal.

Unlike overt political leaning, which can be signaled by direct criticism or praise—identity-related framing is often subtle. It may involve lexical choices, euphemisms, cultural symbols, religious references, variations in narrative emphasis, or omission of contextual information [1,2]. These nuances make identity framing difficult to detect manually, especially at scale, and nearly impossible for computational systems without task-specific resources [11,12].

1.2 Why Framing Matters

Framing influences how audiences make sense of events by guiding attention toward specific aspects of a story [1,3]. It can determine who is blamed, highlight who is victimized, signal moral evaluations, shape public sentiment toward identity groups, and construct long-term narratives that affect social cohesion.

Classic media theory argues that news does not tell people what to think but what to think about and how to think about it [1]. In contexts like Bangladesh, where issues related to religion, culture, and political identity are highly sensitive, framing can amplify tensions or foster understanding [2,9].

However, despite the importance of these dynamics, Bangla NLP currently lacks frameworks or datasets that capture identity-oriented framing patterns [7,8].

1.3 Limitations of Existing Bangla NLP Work

Recent datasets such as BanglaBias and Bangla hyperpartisan classifiers have made progress in identifying political leaning [11, 12], but they do not examine how narratives are shaped within articles. They classify text into categories such as government-leaning, opposition-leaning, or neutral, but do not account for:

- representational imbalance (who gets quoted or centered),
- agency assignment (who is portrayed as active/passive),
- moral framing (e.g., ”good citizen” vs. ”troublemaker”),

- cultural or religious coding (Sanskritized vs. Persianized vocabulary),
- selective contextualization,
- implicit group-level evaluation.

As a result, these resources cannot support large-scale analysis of identity framing in Bangladeshi news—an area with significant socio-political implications [2, 10].

1.4 The Role of LLMs in Media Analysis

Large Language Models (LLMs) such as GPT-4o, Gemini, and Llama-3 offer new possibilities for automated media analysis [13–15]. They can interpret context, perform open-text classification, explain reasoning, and detect subtle patterns when properly prompted.

However, research consistently shows that LLMs can misclassify neutral or ambiguous articles, may reproduce political or cultural biases inherent in their training data, struggle with country-specific sociocultural nuances, and provide inconsistent reasoning on sensitive topics [13, 15].

In Bangladesh specifically, recent evaluations indicate that LLMs sometimes prefer specific ideological sources or misinterpret culturally marked vocabulary [14]. This underscores the need for Bangla-specific framing taxonomies and annotated datasets so LLM outputs can be benchmarked, improved, and audited.

1.5 Problem Statement

Despite the societal relevance of framing and its extensive study in other languages, there is no structured resource for analyzing identity framing in Bengali media. The absence of:

- a conceptual taxonomy,
- a human-labeled dataset,
- evaluation of LLMs on framing tasks,
- a real-time analytical tool,

creates a substantial gap in both academic research and practical media literacy efforts [11, 14].

1.6 Research Purpose

This thesis addresses this gap by developing **NarrativeSense**, a comprehensive, multi-layered framework for analyzing identity framing in Bangla news articles. The project integrates theoretical insights from media studies, linguistic analysis, and computational narratives with practical mechanisms for annotation and LLM evaluation [16, 17].

1.7 Contributions

This thesis makes four core contributions:

1. A theoretically grounded taxonomy of identity framing for Bangla news, built using media theory, sociolinguistics, and domain analysis [1, 3].

2. A high-quality annotated dataset of Bangla news articles capturing multiple framing dimensions at the article level [11, 12].

3. An empirical evaluation of modern LLMs—GPT-4o, Gemini, Llama-3, etc.—on zero-shot, few-shot, and reasoning-based framing detection [13–15].

4. A real-time analytic pipeline design for framing detection in Bangla news streams [17].

Chapter 2

Background

This chapter establishes the theoretical foundations that guide the development of the identity-framing taxonomy proposed in this thesis. While prior research in media bias, political communication, and computational linguistics provides important context, this chapter focuses specifically on *how framing functions*, *how identity is constructed through language*, and *which cultural–linguistic mechanisms shape narrative interpretation in Bangla news*. Drawing insights from sociolinguistics, media studies, discourse analysis, and linguistic anthropology, this chapter provides a comprehensive basis for operationalizing identity framing in Bangladeshi journalism [1,3,4].

2.1 Foundations of Media Framing

2.1.1 Origins: Goffman and Entman

The conceptual roots of framing lie in Erving Goffman's idea of "interpretive schemas" [1], which describes how people use cognitive structures to make sense of events. Building on this, Entman articulated a communication-oriented framing theory, identifying four universal functions:

- **Problem Definition** – identifying what the issue is about.
- **Causal Interpretation** – attributing responsibility or cause.
- **Moral Evaluation** – assigning ethical or moral judgment.
- **Treatment Recommendation** – suggesting policies or solutions.

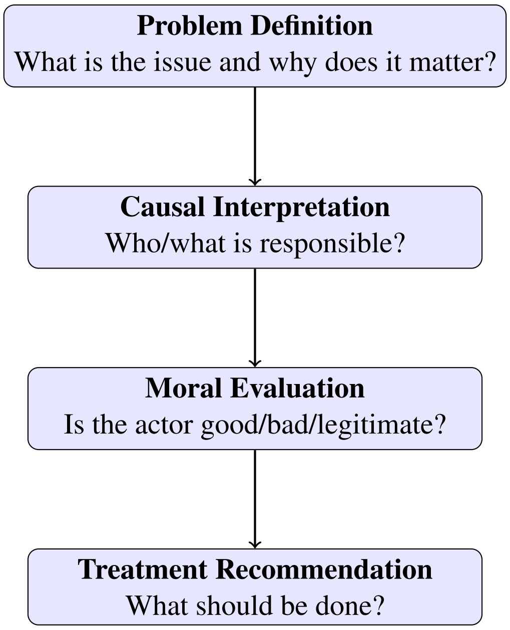

Figure 2.1: Entman's Four Functions of Framing

In journalism, these functions guide how a narrative is shaped—from headline construction to source selection—ultimately shaping audience perception. These theoretical principles are foundational for understanding how identity groups are portrayed, valorized, or stigmatized in Bangla news [14, 17].

2.1.2 Agenda-Setting and Gatekeeping

Agenda-setting theory posits that media influences not only public knowledge but also the importance assigned to various issues [18, 19]. Editors and newsrooms act as gatekeepers, prioritizing specific events, actors, and interpretations. In the Bangladeshi context:

- political events, protests, and institutional statements often dominate the news cycle,
- whereas minority issues, local conflicts, and systemic failures receive comparatively less coverage.

These selective decisions shape Selection Bias. Understanding this bias is essential, because framing cannot occur without prior selection. The uneven visibility of certain groups or issues is often the first step in constructing a narrative that reinforces ideological or identity-based boundaries [4, 10].

2.1.3 Selection Bias vs. Framing Bias

Media bias operates through two connected mechanisms:

- **Selection Bias** – which events are emphasized or ignored.
- **Framing Bias** – how selected events are described, interpreted, or linguistically shaped.

Example (Selection Bias): Suppose two Bangladeshi news outlets cover the same day's events. One outlet gives continuous coverage of a garment-workers' protest in Gazipur. Another outlet ignores it and highlights entertainment news instead.

A realistic contrasting example:

- **Outlet A headline:** গাজীপুরে শ্রমিকদের বিক্ষোভ, সড়ক অবরোধে উত্তেজনা
- **Outlet B headline:** নতুন ছবির ঘোষণা দিলেন জনপ্রিয় তারকা

Example (Framing Bias): Two outlets may report the same event but apply different tones:

- **Outlet A framing:** অরাজক পরিস্থিতি তৈরি করেছে শ্রমিকরা
- **Outlet B framing:** ন্যায্য মজুরির দাবিতে শ্রমিকদের শান্তিপূর্ণ বিক্ষোভ

Framing bias is harder to quantify because there is no single correct way to describe an event [20, 21].

2.2 Identity Construction Through Language

Identity framing is not merely a matter of describing events; it involves deeper mechanisms through which linguistic choices shape public perception of groups or individuals.

2.2.1 Social Actor Representation (van Leeuwen)

Theo van Leeuwen's framework identifies key strategies used to represent people in discourse:

- **Activation vs. Passivation** – portraying groups as active agents or passive recipients.
- **Personalization vs. Impersonalization** – assigning human traits or institutional abstraction.

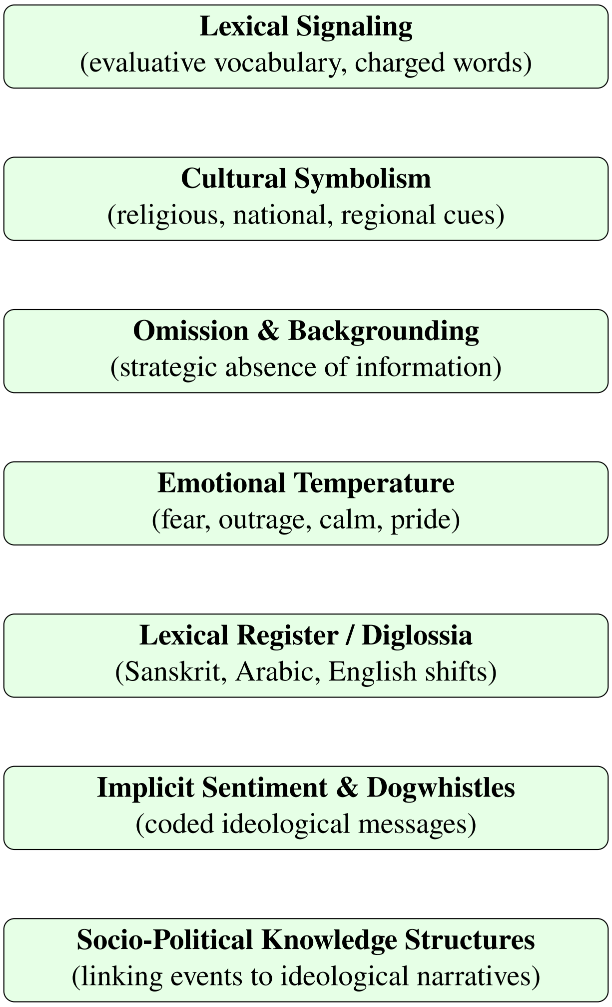

Figure 2.2: Layers of Identity Construction in Bangla News Narratives

- **Genericization vs. Specification** – describing people as categories or as specific individuals.

These strategies influence whether groups appear threatening, victimized, marginalized, or empowered [3, 22]. Bangla news articles frequently rely on these techniques—sometimes overtly, sometimes implicitly—to convey political or cultural undertones.

2.2.2 Bangla Linguistic Signaling

Bangla exhibits rich lexical diversity influenced by Sanskrit, Perso-Arabic, and English [10, 23]. These lexical registers serve as framing signals:

- **Sanskritized vocabulary** often indexes nationalism, secularism, or formality.
- **Perso-Arabic terms** carry religious, communal, or traditional associations.
- **English borrowings** evoke institutional authority or bureaucratic neutrality.

For example:

- "অন্ত্যেষ্টিক্রিয়া" – ceremonial, cultural.
- "জানাজা" – religious, Muslim identity.

These lexical choices play a crucial role in constructing identity frames that transcend literal meaning [16, 24].

2.2.3 Dogwhistles and Coded Language

Certain Bangla terms function as dogwhistles—seemingly neutral expressions that carry coded ideological signals [14, 25]. Examples include:

- "উগ্রপন্থী" – implies extremist violence without specifying ideology.
- "দুর্বৃত্ত" – neutralizes perpetrator identity through generic labeling.

2.2.4 Syntactic Framing

Bangla syntax allows manipulation of agency and blame [22]:

- "লোকটি মারা গেছে" – natural, event-focused.
- "লোকটিকে মারা হয়েছে" – implies external violence.

2.3 Narrative Layers in Bangla News

Identity framing frequently emerges from the interaction of multiple narrative layers [5, 11]:

1. **Lexical signaling** – charged or evaluative vocabulary.
2. **Cultural symbolism** – references to religious, historical, or national icons.

3. **Omission and backgrounding** – strategic exclusion of key details.

4. **Emotional temperature** – inducing calm, fear, outrage, or pride.

5. **Lexical register (diglossia)** – Sanskrit vs. Arabic vs. English code preference.

6. **Implicit sentiment and dogwhistles** – indirect value judgments.

7. **Socio-political knowledge structures** – linking events to broader ideological narratives.

Each layer contributes to constructing a framing environment in which identity groups are evaluated, ranked, or positioned within public discourse.

2.4 Operationalizing Theory for Annotation

The theoretical insights discussed above support all components of this research pipeline:

- **Taxonomy Development** – selecting framing categories grounded in linguistic and cultural analysis [26].
- **Annotation Protocol** – designing instructions that capture implicit and explicit identity cues [15].
- **LLM Evaluation** – testing zero-shot and few-shot capabilities using theory-informed prompts [19].
- **Framework Integration** – enabling NarrativeSense to interpret identity frames in real-world news [8].

By grounding the annotation system in established theories of framing and discourse, the resulting taxonomy achieves both conceptual robustness and cultural relevance for Bangla media.

Chapter 3

Related Work

This chapter reviews prior research across political bias detection, news framing studies, LLM-based media analysis, and Bangla NLP. It synthesizes contributions from classical framing corpora, multilingual framing expansions, LLM-powered detection tools, and recent South Asian media research. Together, these works establish the global foundation for computational media analysis while highlighting the absence of identity-framing resources for Bangla [3, 11, 27].

3.1 Computational Media Bias, Stance, and Framing Detection

3.1.1 Framing and Media Bias Corpora

One of the earliest large-scale framing datasets is the **Media Frames Corpus (MFC)** introduced by Card et al. [3]. It demonstrated that issue framing—problem definition, moral evaluation, causal attribution, and policy recommendation—can be systematically annotated and modeled using NLP. Subsequent expansions, such as **The Media Frames Corpus Across Issues** and multilingual variants [6,27], extend framing annotations across languages, topics, and sociopolitical contexts. These resources confirm that framing is quantifiable, but are overwhelmingly centered on Western issues such as immigration, gun control, and public health [4, 20].

Recent domain-specific framing research—including *Detecting Frames in News Headlines* [4], *Framing Trends in Gun Violence* [4], and *Automatic Temporal Framing Detection* [28]—has explored how frames evolve over time and across events. These works emphasize:

- frame shifts during crises,

- event-centric framing,
- temporal variance in political narratives,
- the role of headlines as framing devices.

Limitation: Although these studies demonstrate the feasibility of large-scale framing annotation, none address identity framing, and none include Bangla-specific sociolinguistic cues [2].

3.1.2 Bias and Stance Detection in News Media

Beyond framing, bias detection tools have emerged to identify ideological leaning, source-level bias, and stylistic cues [29–31]. Notable contributions include:

- **BiasScanner (2024)** — browser-integrated real-time sentence-level bias detection [17],
- **Political Lean Detection Models** — analyses built on BASIL, AllSides, and syntactic/lexical ideological features [21, 32],
- **Selection vs. Framing Bias Studies** — exploring how media organizations systematically highlight or obscure particular actors or events [1, 33].

These studies collectively demonstrate that media bias is computationally measurable, but they operate primarily on English-language corpora and focus on political leaning rather than the deeper identity-framing structures.

3.2 LLMs for Framing, Credibility, and Bias Analysis

3.2.1 LLMs for Narrative Interpretation and Framing Detection

Recent work such as **Decoding News Narratives** [13] and **A Critical Analysis of LLMs in Framing Detection** [13] evaluates GPT-4, LLaMA-3 [34], Mistral, and other LLMs on the task of identifying frames in news articles. Findings consistently show:

- LLMs excel with explicit frames but struggle with implicit cultural cues,
- identity-related frames are misinterpreted without contextual grounding,
- chain-of-thought prompting improves accuracy but increases hallucination risks [35].

3.2.2 LLM Annotation: Promises and Risks

The Promises and Pitfalls of LLM Annotations in Dataset Labeling [15] demonstrates that LLM-generated labels can significantly accelerate corpus development but remain susceptible to:

- political bias,
- semantic hallucination,
- inconsistent application of frame taxonomy,
- cultural misinterpretation in low-resource languages.

This reinforces the need for human-in-the-loop verification, especially for Bangla's culturally encoded discourse.

3.2.3 Credibility and Bias Evaluation in Bangladesh

The study **Evaluating Credibility and Political Bias in LLMs for News Outlets in Bangladesh** [14] assesses how LLMs judge credibility and ideological leaning of Bangladeshi media sources. Key insights:

- models show systematic preference toward left-aligned sources,
- LLM credibility judgments correlate with their Western training biases,
- LLMs misread culturally marked Bangla expressions, leading to incorrect bias attribution.

Implication: LLMs require culturally grounded resources to accurately interpret Bangla news narratives [25].

3.3 Real-Time Media Bias Detection and HCI Approaches

3.3.1 Media Bias Detector System

The **Media Bias Detector** [17] presents a full-stack system for real-time selection and framing bias analysis. The system:

- ingests news articles continuously,

- uses LLMs to classify tone, lean, and frame,
- identifies disparities in event coverage across publishers,
- combines automated annotation with human oversight.

This framework demonstrates that LLM-assisted analysis can scale to daily media monitoring but also confirms that full automation is unreliable without cultural grounding [36].

Relevance: The system directly inspires the architecture and evaluation design for NarrativeSense but lacks identity-framing dimensions and Bangla linguistic markers.

3.4 Bangla and South Asian Media Analysis

3.4.1 Political Bias in Bangla News

Read Between the Lines: A Benchmark for Uncovering Political Bias in Bangla News Articles [11] created the first structured benchmark for Bangla political stance. It introduced:

- a three-class political leaning schema,
- 200 manually annotated political articles,
- evaluation of 28 LLMs and transformer baselines.

Results show strong performance on explicit critique but reveal:

- persistent confusion in the “neutral” class,
- overprediction of political bias in fact-based articles,
- inability to correctly interpret moderate or ambiguous narratives.

3.4.2 Framing in South Asian Media

Social-science framing studies such as:

- *News Framing in Bangladesh, India and British Media: Bangladesh Parliamentary Election 2018* [2],

- studies on communal conflict framing and political rhetoric [9, 23],
- research on institutional portrayal and media polarization [5, 10],

show the deep cultural embedding of identity signaling in South Asian journalism. However, these works are **qualitative** and lack computational taxonomies or datasets.

3.5 Summary of Gaps

Overall, existing research provides strong precedents for computational analysis of framing and bias but leaves three critical gaps:

1. no Bangla **identity-framing taxonomy or dataset** [2, 11],
2. no evaluation of LLMs on **identity framing** [14, 25],
3. no integrated framework for real-time analysis of Bangla media narratives [17].

This thesis directly addresses these gaps by developing a comprehensive identity-framing resource, annotation framework, and LLM benchmark for Bangla media.

Chapter 4

Research Gap & Objective

This chapter synthesizes insights from contemporary research in media bias, computational linguistics, and Bangla news studies to articulate the precise gap that this thesis addresses. While previous work has advanced political bias detection and stance classification, these models and datasets capture only a narrow portion of how narratives are constructed in Bangla journalism. Identity framing—how social groups, political actors, institutions, and communities are linguistically portrayed—remains understudied, theoretically complex, and computationally underrepresented. This chapter explains why identity framing is analytically essential, how it differs from political leaning, why it remains technically challenging, and what motivates the development of a comprehensive Bangla framing benchmark.

4.1 Why Identity Framing Matters in Bangla Media

The Bangladeshi media landscape is deeply interwoven with political contestation, communal sensitivities, cultural hierarchies, and historical narratives. As a result, identity-relevant cues are frequently embedded within news articles, often in subtle and culturally encoded forms. These cues may appear through lexical register differences (such as shifts between Sanskritized and Perso-Arabic vocabulary), selective emphasis on identity markers, portrayal of groups as responsible or victimized, symbolic language, or dog-whistle terms used to imply threat, illegitimacy, or moral judgment.

Identity framing shapes how readers interpret social conflict, protest movements, communal tension, political legitimacy, institutional failures, and even ordinary social events. A slight linguistic adjustment—such as describing a group as ``উগ্রপন্থী'' ("extremist") rather than ``কর্মী'' ("activist")—can shift moral valence and reposition a community within a narrative hierarchy. These representational patterns influence public perception and help construct broader ideological and social boundaries.

Despite the centrality of identity in Bangladeshi public discourse, no computational resource currently captures how newspapers frame actors or communities. Existing systems cannot interpret cultural, linguistic, or symbolic signals that underlie identity construction, making automated analysis incomplete and potentially misleading.

4.2 Framing Bias vs. Political Lean: Two Orthogonal Axes

Political lean reflects alignment toward political actors such as the ruling party or the opposition. Identity framing, however, constructs symbolic and cultural boundaries based on religion, ethnicity, class, region, occupation, or institutional affiliation. Unlike political lean, identity framing frequently shifts with topic, crisis, or strategic communicative need.

Most Bangla NLP research to date has focused on political leaning—typically a three-way classification of government-leaning, opposition-leaning, or neutral. While such labels are useful for ideological categorization, they cannot explain how identities are framed or how narratives are shaped within a story. Political leaning is largely stable across issues; identity framing is fluid, strategic, and context-dependent.

Political stance alone cannot capture how:

- communities are described,
- agency or blame is assigned,
- moral or cultural frames are activated,
- contextual details are selectively emphasized or omitted,
- events are narrated differently depending on actors involved.

Identity framing, therefore, forms a separate analytical axis. Political lean may signal *who* the outlet favors; identity framing reveals *how* the outlet constructs narratives around social groups.

The two dimensions frequently diverge. For example, a pro-government publication may portray youths as “entrepreneurial and forward-looking’’ while simultaneously depicting student organizations as “agitators destabilizing campuses.’’

Identity framing thus varies across communities, issues, and contexts—even within the same publication—proving that ideological stance and identity portrayal must be measured independently.

Table 4.1 illustrates how a single political actor (e.g., the ruling party) may adopt multiple and sometimes contradictory identity frames depending on context:

Political Lean	**Identity Framing**	**Example**
Pro-Government	Positive to Youths	সরকারের উদ্যোগে গ্রামের যুবসমাজ উদ্যোক্তা হিসেবে এগিয়ে আসছে।
Pro-Government	Negative to Student Orgs.	সাম্প্রতিক ভাঙচুরের ঘটনায় ছাত্র সংগঠনগুলো পরিস্থিতি উত্তপ্ত করার চেষ্টা করছে।
Anti-Government	Positive to Jhum Farmers	চট্টগ্রামের জুমচাষি সম্প্রদায় সংকট কাটাতে নিজেরাই সৃজনশীল সমাধান বের করেছে, কিন্তু সরকারি সহায়তা পাচ্ছে না।
Anti-Government	Negative to Contractors	সরকারের নীতির সুবিধা নিয়ে শহরের ঠিকাদার গোষ্ঠী অতিরিক্ত লাভ তুলছে এবং জনগণ ভোগান্তিতে পড়ছে।

Table 4.1: Divergence of Political Lean and Identity Framing

This divergence demonstrates why identity framing cannot be approximated using political leaning labels alone.

4.3 What Current Bangla Benchmarks Cannot Capture

Existing Bangla datasets—including those targeting political ideology or hyperpartisanship—capture surface-level markers of bias but fail to represent underlying narrative mechanisms such as responsibility framing, conflict framing, or group portrayal. These datasets lack annotations for representational strategies, symbolic cues, and issue-specific frames that international corpora like the Media Frames Corpus (MFC) or BASIL provide.

Moreover, Bangla framing cues rely heavily on culturally loaded lexical patterns: Sanskritized vocabulary often indexes nationalist or secular frames, Perso-Arabic vocabulary often indexes religious or communal frames, and English borrowings index institutional or bureaucratic frames.

None of these linguistic registers are encoded in existing datasets or accessible to current models.

Equally important is the lack of frame-level annotations. Without annotated data capturing the portrayal of actors, narrative emphasis, and contextual omissions, computational systems cannot interpret how Bangladeshi news constructs meaning.

4.4 Why Identity Framing Is Computationally Challenging

Identity framing is inherently multidimensional and often implicit. Signals may be distributed across sentences or embedded in metaphor, cultural reference, or syntactic manipulation. Understanding such cues requires culturally grounded knowledge—of Bangladeshi politics, religious sensitivities, regional dynamics, linguistic registers, and symbolic expression.

Further challenges arise from:

- the absence of objective ground truth,
- the overlap of multiple frames in a single article,
- the difficulty of distinguishing neutrality from subtle bias,
- the need for multi-label classification,
- the requirement for adjudicated annotation.

Thus, automated identity framing analysis demands a robust codebook, consistent annotation protocol, and multilayered modeling design.

4.5 Insufficiency of Existing LLM Tools for Bangla Framing

Although recent work evaluates LLM performance on political bias, credibility, and stance detection, no study tests State-of-the-Art LLMs on Bangla framing tasks. Models such as GPT-4o, Gemini, or Llama-3 exhibit prompt sensitivity, inconsistent predictions across repeated queries, and hallucinations when interpreting Bangla cultural or symbolic expressions. Furthermore, these models often impose Western-centric political frames onto Bangla news, leading to “AI-induced framing bias.’’

Without a dedicated evaluation benchmark, such errors cannot be measured or mitigated.

4.6 The Research Gap

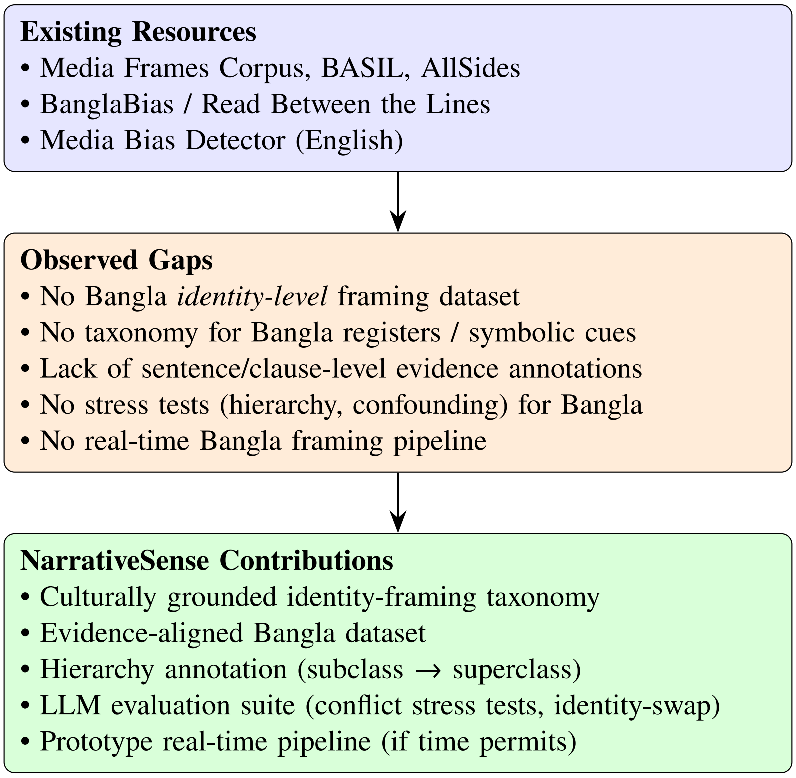

Figure 4.1: Compact vertical visual gap analysis illustrating how NarrativeSense fills the gaps in existing Bangla framing research.

Despite progress in political bias detection, three fundamental gaps persist:

1. Absence of Bangla Frame-Level Annotation

No dataset captures issue framing, responsibility assignment, moral narratives, or portrayals of identity groups. As a result, Bangla NLP lacks the resources to study framing in a systematic manner.

2. Lack of LLM Evaluation for Framing Tasks

No comparative study examines the zero-shot, few-shot, or chain-of-thought performance of modern LLMs on Bangla framing, nor their alignment with human annotation.

3. No Framework or tool for Real-Time Analysis

Bangladesh lacks a computational system analogous to the Media Bias Detector for English. There is no architecture for real-time ingestion, multi-frame detection, or visualization of narrative shifts across publishers.

These gaps collectively prevent the development of transparent media monitoring tools and limit academic inquiry into Bangla discourse.

4.7 Research Objectives

To address the identified gaps, this study aims to develop a comprehensive, multi-stage framework for Bangla news framing analysis, utilizing both human annotation and LLM-based evaluation.

Objective 1: Development of a Bangla Framing Taxonomy and Codebook.

We aim to construct an empirically grounded framing schema tailored to the socio-political, cultural, and media-specific context of Bangladesh. This involves incorporating insights from international literature (MFC, BASIL) while ensuring regional relevance, and establishing a vetted, multi-label annotation protocol covering issue frames, actor portrayal, responsibility, and narrative tone.

Objective 2: Dataset Creation & Human-Labeled Benchmark.

The study will curate a balanced corpus of Bangla news articles across diverse publishers and topics. Through multi-stage annotation with rigorous inter-annotator agreement evaluation, we will build the first Bangla Framing Detection Benchmark to enable model comparison and future research.

Objective 3: Performance Evaluation of LLMs on Frame Detection.

We will evaluate leading LLMs (GPT-4o, Gemini, Llama-3, etc.) using zero-shot classification, few-shot in-context learning, and chain-of-thought prompting. The goal is to assess model accuracy, consistency, and investigate whether LLMs introduce ”AI-induced framing bias” when processing low-resource languages like Bangla.

Objective 4: Computational Architecture for Real-Time Analysis.

Finally, we aim to develop a practical pipeline architecture that can ingest Bangla news in real-time, detect multiple framing dimensions, analyze publisher-level differences, and visualize temporal narrative shifts, ensuring the system aligns with HCI principles of interpretability and transparency.

Chapter 5

Methodology

This chapter presents the end-to-end methodological framework used to construct the NarrativeSense dataset, annotate identity framing in Bangla news, and evaluate Large Language Models (LLMs). The methodology integrates theory-driven identity definitions, event-based corpus design, a multi-level annotation protocol, and a computational modeling pipeline tailored to the linguistic and cultural nuances of Bangla. The chapter is organized into four major components:

1. Data Collection and Identity–Event Corpus Construction
2. Annotation Schema and Human-Centered Labeling Framework
3. Computational Modeling and LLM Evaluation Pipeline
4. Validation, Bias Testing, and Reliability Assessment

5.1 Data Collection and Corpus Design

Media-driven identity framing rarely manifests in a vacuum; it becomes visible when group identities intersect with socio-political friction, institutional actions, or civil crises. To capture this dynamic comprehensively, our data collection architecture operates as an iterative funnel—transitioning from high-volume unstructured web harvesting to a highly curated, event-clustered dataset.

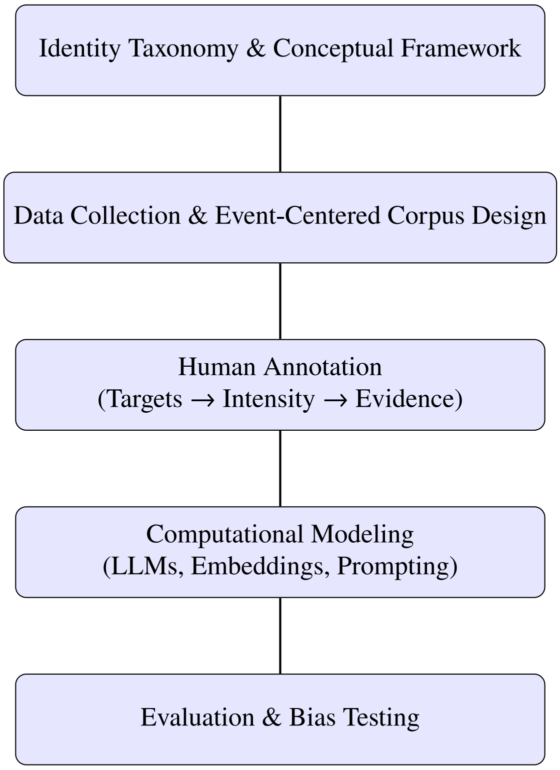

Figure 5.1: High-level Research Framework / Pipeline Overview

5.1.1 Empirically Grounded Identity Taxonomy

Rather than relying on abstract, classical demographic divisions, the conceptual framework of this study is built upon an empirically grounded taxonomy. Following an exploratory analysis of the raw scraped text, we operationalized a strict, multi-layered schema of code keywords that actively function as identity markers within contemporary Bangladeshi media discourse.

Crucially, this taxonomy was designed to capture both intrinsic identities (e.g., ethnicity, religion) and situational, conflict-driven identities (e.g., protesters, victims, geopolitical actors). The resulting lexical matrices group these operational identities into five structural dimensions:

- **Ethno-Religious and Demographic Boundaries:** Lexical terms distinguishing religious majorities, minority communities, and historically marginalized or displaced demographic groups.
  - *Religious Entities:* Encompassing mainstream and minority faith groups (e.g., মুসলিম, হিন্দু, সংখ্যালঘু, আহমদিয়া, বাউল).

- *Ethno-Demographic Groups:* Indigenous and refugee populations (e.g., পাহাড়ি, আদিবাসী, রোহিঙ্গা).

- **Political, Ideological, and Partisan Affiliations:** Distinct partisan factions and ideological divides that drive media polarization.
  - *Partisan Formations:* Mainstream and grassroots political entities (e.g., "আওয়ামী লীগ", "বিএনপি", "জামায়াতে ইসলামি", "বামপন্থী").
  - *Ideological Divides:* Terms isolating secularists from conservative factions (e.g., প্রগতিশীল, সুশীল, নাস্তিক versus হেফাজতে, কওমি, আলেম).
- **State Apparatus and Institutional Actors:** Institutional bodies and personnel that enforce, mediate, or contest systemic power.
  - *Law Enforcement & Military:* (e.g., "পুলিশ", "সেনাবাহিনী", "র‍্যাব").
  - *Administrative & Judicial:* (e.g., "সরকার", "আদালত", "দুদক").
- **Socio-Economic, Professional, and Civic Cohorts:** Functional demographic groups frequently mobilized in mass protests, economic friction, or civil society debates.
  - *Mobilized Cohorts:* Students, laborers, and vulnerable groups (e.g., শিক্ষার্থী, শ্রমিক, বস্তিবাসী).
  - *Professional & Elite Classes:* (e.g., সাংবাদিক, বুদ্ধিজীবী, সিন্ডিকেট, ব্যবসায়ী).
- **Issue-Specific Framings and Geopolitical Entities:** This dimension captures situational identities that emerge explicitly during flashpoints, as well as external geopolitical actors heavily referenced in domestic political framing.
  - *Conflict & Action Roles:* Ephemeral but highly charged identities born from events (e.g., "বৈষম্যবিরোধী □□□□□ আন্দোলনকারী", "বিক্ষোভকারী", "ভুক্তভোগী").
  - *Pejorative/Extremist Framing:* A critical methodological distinction separating generic religious identity from militant or mob-based framing (e.g., উগ্রপন্থী, মৌলবাদী, জঙ্গি, মব).
  - *Geopolitical Influencers:* External states acting as ideological or political entities in local narratives (e.g., ভারত, বিএসএফ, আমেরিকা, স্যাঙ্কশন, জাতিসংঘ).

By computationally separating generic descriptors (e.g., *Muslim*) from highly charged ideological framings (e.g., *Conservative Islamist* or *Extremist Mob*), this data-enriched taxonomy ensures the retrieval pipeline is grounded in the actual linguistic reality of the media ecosystem. It allows the research to systematically track not just *who* is being reported on, but *how* their identity is linguistically constructed during crises.

5.1.2 The Iterative Retrieval and Curation Pipeline

To successfully isolate articles demonstrating robust identity framing, our methodology employs a sequential four-phase extraction pipeline:

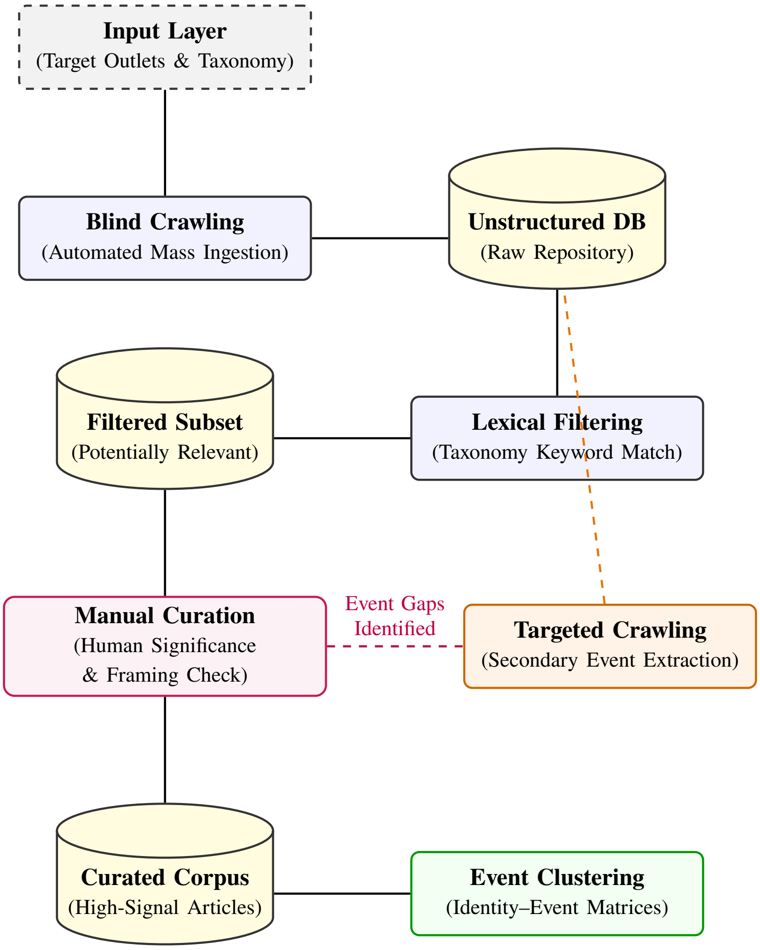

Figure 5.2: Architectural flow of the Iterative Purposive Sampling Pipeline.

5.1.2.1 Phase 1: Unstructured Stream Harvesting (Blind Crawling)

Because identity narratives are distributed unpredictably across news categories, the ingestion pipeline begins with automated, unstructured scraping. Custom Python-based crawlers execute continuous, broad-spectrum ingestion of major Bangla news outlets. This initial ”blind” crawl harvests a massive raw repository without applying premature semantic constraints, ensuring no peripheral contextual coverage is missed.

5.1.2.2 Phase 2: Lexical Filtering via Taxonomy

The massive raw dataset is subsequently parsed and programmatically filtered using the established identity taxonomy. By applying targeted keyword-matching matrices utilizing the lexical identity terms outlined above, this phase algorithmically culls the unstructured dataset down to an intermediate corpus containing articles where targeted socio-political identities are explicitly mentioned.

5.1.2.3 Phase 3: Purposive Manual Curation

Because the mere lexical presence of an identity keyword does not guarantee active *framing*, the filtered subset undergoes human-guided exploration. Annotators manually review the intermediate corpus to isolate significant articles that genuinely fit the research parameters. This ensures the selection of texts that construct active narratives around these identities through out-group stereotyping, political mobilization, or symbolic controversies.

5.1.2.4 Phase 4: Targeted Extraction and Event-Centered Clustering

Following manual selection, the identified high-impact incidents guide a secondary round of targeted crawling. This focused ingestion captures all surrounding context and cross-platform coverage of the isolated events. Finally, the finalized dataset is structurally organized into *Identity–Event Clusters*. Grouping these articles around specific ”Flashpoint Events” allows the research to systematically compare how different media publishers frame the exact same socio-political identity within the bounds of a shared crisis.

5.2 News Article Annotation Strategy

To systematically deconstruct the underlying linguistic and ideological mechanisms of media framing, we introduced a specialized annotation infrastructure. Rather than assigning superficial, flat category labels to whole articles, annotators are guided through a granular textual dissection process. The annotation schema follows a strict three-tier hierarchical analytical flow.

5.2.1 Annotation Schema

5.2.1.1 Step 1: Identifying Targets

Annotators first identify the primary targets present in the text based on the Identity Targets table (defined in Section 4.1). An article is deemed valid for the dataset only if a target identity acts as the protagonist or antagonist of the narrative.

Table 5.1 shows the 5-point Likert scale used for entity framing intensity scoring, capturing nuances beyond simple sentiment analysis.

5.2.1.2 Step 2: Framing Intensity Scoring (5-Point Likert Scale)

For each identified entity, the annotator determines the framing stance. Unlike binary sentiment (Good/Bad), we utilize a weighted 5-point intensity scale to capture the nuance between "criticism" and "demonization," or "praise" and "glorification" [11].

Score	Label	Definition
+2	Highly Positive	Glorification or heroic portrayal; morally elevated identity representation.
+1	Positive	Supportive or approving portrayal.
0	Neutral	Factual reporting without emotive cues or bias.
-1	Negative	Critical or unfavorable portrayal.
-2	Extremely Negative	Vilification, demonization, or portrayal as a societal threat.

Table 5.1: 5-Point Likert Scale for Framing Intensity Scoring of Entities

5.2.1.3 Step 3: Evidence-Based Narrative Construction

Beyond assigning labels, annotators must identify the specific linguistic evidence responsible for framing. This step ensures interpretability and enables fine-grained evaluation of model reasoning. Evidence may appear at different levels of granularity, including full sentences, clauses, or key lexical items. **Examples of Valid Evidence Types:**

- **Sentences:** Complete sentences that explicitly assign blame, praise, victimhood, or moral judgment.
  - "পুলিশ গুলি করে অন্তত তিনজনকে হত্যা করে"
  - "সরকারের সিদ্ধান্তে জনমনে স্বস্তি ফিরে এসেছে"
  - "শিক্ষার্থীদের বেপরোয়া আচরণে ক্যাম্পাস অস্থিতিশীল হয়ে পড়ে"
- **Clauses:** Partial sentences that contain moral, emotional, or agentive framing.
  - "...যা সাধারণ মানুষের মধ্যে আতঙ্ক ছড়িয়ে দেয়" (Clause framing an identity as causing public fear.)
  - "...যদিও ভুক্তভোগীরা দাবি করেন পুলিশ যথাসময়ে আসেনি" (Implicates institutional negligence.)
  - "...যেখানে ওই গোষ্ঠীকে 'উগ্র' বলে উল্লেখ করা হয়" (Clause with explicit labeling.)
- **Key words/phrases:** Highly charged lexical items that signal identity framing, such as "উগ্র" [radical], "সন্ত্রাসী" [terrorist], "অসহায়" [helpless], "ধর্মীয় অনুভূতি" [religious sentiment], "কোটা আন্দোলনকারি" [quota protester], "দুষ্কৃতকারী" [miscreant], "দেশবিরোধী" [anti-national].

This multi-level evidence extraction creates an "evidence-aligned" benchmark similar to FEVER-style fact verification pipelines. It ensures that framing decisions are grounded in textual reality and allows the evaluation of whether LLMs can detect not only *what* the frame is, but also *why* a frame has been assigned. These snippets become part of an evidence-aligned benchmark similar to FEVER-style fact-checking datasets. This is critical for evaluating whether LLMs can reason about narrative cues, not just classify sentiment.

5.2.2 Human Annotation Strategy

The annotation of identity framing requires careful human judgment, as the cues are often subtle and embedded within linguistic patterns, sourcing choices, and narrative emphasis. Our annotators do not only label sentiment; they evaluate how a narrative is constructed around socio-political identities.

5.2.2.1 Parallel Annotation

Each article is annotated independently by two expert annotators using the full identity-based schema described earlier (Target Identification → Framing Intensity Scoring → Evidence Extraction). Annotators are instructed to examine both explicit cues (adjectives, blame attribution, praise markers) and implicit framing signals, such as:

- Source selection (e.g., only quoting police, only quoting victims).
- Foregrounding or backgrounding specific identities.
- Contextual juxtapositions (e.g., linking an identity with crime, virtue, victimhood).
- Narrative direction (who is centered, who is backgrounded, who is portrayed as responsible).

This ensures that annotation reflects not just surface-level sentiment but deeper narrative structures.

5.2.2.2 Adjudication

Disagreements between Annotator 1 and Annotator 2 are resolved by a third senior annotator who acts as the adjudicator. The adjudicator carefully re-evaluates:

- Identity detection accuracy.
- Consistency of framing intensity scores.
- Correctness of evidence selection.
- Whether the narrative portrayal matches the assigned stance.

The adjudicated labels become the final gold-standard annotations for the dataset.

5.2.2.3 Inter-Annotator Reliability

We measure the consistency between the two primary annotators using Fleiss' Kappa, which provides a statistical estimate of agreement beyond chance. Most disagreements typically arise between:

- Neutral vs. Mildly Positive/Negative framing.
- Positive vs. Highly Positive.
- Negative vs. Extremely Negative.

This pattern reflects the inherent ambiguity in narrative framing, where subtle linguistic cues may shift interpretation.

5.2.2.4 Annotation Philosophy

Our annotation design emphasizes:

- **Interpretability:** Annotators must justify their stance scores with explicit textual evidence.
- **Identity Precision:** Labels are tied to clearly defined identity categories, minimizing subjective drift.
- **Event Sensitivity:** Annotators consider the social and historical context of each event cluster to detect implicit biases.

This structure ensures that annotations capture not merely what the article says, but how it constructs socio-political identities and their roles in the narrative. Moreover, this annotation workflow can be extended easily. New socio-political identities, future events, or additional news sources can be incorporated without changing the core structure, making the pipeline adaptable for long-term use. Overall, this human-centered annotation strategy forms the foundation of NarrativeSense. It allows the benchmark to capture how Bengali news constructs, elevates, marginalizes, or weaponizes identities, enabling precise evaluation of LLM performance on fine-grained identity framing.

5.3 Computational Modeling

5.3.1 Model Architecture Selection

We will evaluate a tiered hierarchy of models to determine the optimal balance between performance and resource efficiency.

1. **Tier 1: Proprietary Giants (GPT-4o, Claude 3.5 Sonnet):** These models serve as the benchmark for reasoning capabilities. They will be used via API with Few-Shot Chain-of-Thought (CoT) prompting to establish a performance ceiling.

2. **Tier 2: Open-Source Massive Models (Llama-3-70B, Qwen-2.5-72B):** These models have shown strong multilingual performance and can be hosted locally, addressing data privacy concerns.

3. **Tier 3: Domain-Specific Compact Models (BanglaBERT, Fine-tuned Mistral):** We will fine-tune smaller models (7B parameters) specifically on our Gold Standard dataset. The hypothesis is that a smaller, domain-adapted model may outperform a larger, generalist model in detecting local nuance.

4. **Tier 4: Lightweight Small-Scale Models (2B–4B parameter models):** This tier includes extremely lightweight architectures such as Llama-3-3B, Qwen-2.5-3B, Mistral 3B, and similar models optimized for on-device or resource-constrained environments. These models are evaluated to explore:

   - whether identity framing can be detected with minimal computational cost,
   - how performance scales down in low-parameter settings,
   - the feasibility of integrating framing detection into mobile or real-time newsroom tools.

5.3.2 Prompt Engineering & Chain-of-Thought (CoT) Protocols

5.3.2.1 Zero-shot vs. Few-shot

We evaluate in three modes for each model:

- **Zero-shot:** Instruction only.

- **Few-shot CoT:** 2–8 in-context exemplars demonstrating entity extraction, score assignment, and evidence extraction.

5.3.2.2 CoT Prompt Design

For few-shot CoT, we create compact exemplars that include:

- Small article snippet.
- Annotator-style reasoning steps (Identify entities → Lexical cues → Sourcing → Frame determination).
- Final structured answer (entity:score:evidence).

5.4 Bias and Robustness Testing

We incorporate several robustness strategies:

5.4.1 Identity-Swap Test

Identity terms are swapped (e.g., Hindu Muslim) to detect model-inherited stereotyping.

5.4.2 Neutral-Class Stability Test

Ensures models avoid collapsing ambiguous reporting into biased categories.

5.4.3 Prompt Variation Stability

Tests sensitivity to prompt paraphrasing and instruction structure.

5.5 Evaluation and Validation Metrics

We use:

- F1 (micro, macro),
- accuracy,

- multi-label Jaccard,
- reasoning consistency,
- qualitative narrative error analysis.

5.6 Summary

This chapter presented the complete methodological pipeline for constructing the NarrativeSense dataset and evaluating LLMs on Bangla identity framing. The methodology integrates theoretical grounding, identity-guided corpus design, structured annotation, low-resource-aware modeling, and rigorous robustness testing.

Chapter 6

Future Work

As this thesis is currently in the middle phase of development, several major research components are still in progress. The pilot study has provided crucial diagnostic insights—particularly regarding hierarchical identity failures in GPT-5 and thematic confounding in Gemini 3 Pro—that now inform a refined roadmap for the remaining stages of the NarrativeSense framework. Future work will proceed in alignment with the four core research objectives outlined in Chapter 4, with the understanding that Objective 4 will be pursued only if time permits.

6.0.1 Expansion of the Corpus and Completion of the Framing Taxonomy (Objective 1)

The next phase involves expanding the Bangla framing taxonomy and finalizing the multi-label codebook. Specifically:

- enriching identity categories to better capture hierarchical relations (e.g., subclass–superclass),
- incorporating culturally sensitive markers such as register variation, honorific structures, Sanskritized vs. Perso-Arabic vocabulary, and implicit symbolic language,
- refining guidelines to prevent the interpretive drift that affects both annotators and LLMs.

The pilot study's findings on identity hierarchy confusion directly motivate improved schema rules to ensure that future annotations support robust model learning.

6.0.2 Dataset Construction, Annotation Calibration, and Reliability Validation (Objective 2)

With the feasibility testing complete, the next major task is to continue building the large-scale Bangla Framing Benchmark. This includes:

- expanding the corpus to include diverse publishers, regions, and event types,
- conducting multi-round annotator training and calibration based on early failure cases,
- implementing an adjudication workflow that explicitly resolves framing ambiguity,
- measuring inter-annotator agreement using Fleiss' Kappa and iteratively refining guidelines.

Pilot errors such as thematic confounding confirm the necessity of strong evidence-based annotation rules capable of disambiguating victimhood, villainy, and contextual negativity.

6.0.3 Full-Scale Evaluation of LLMs with Stress Tests (Objective 3)

The next stage will extend beyond minimal prompting and run a structured evaluation across major LLMs (GPT-5, Gemini 3 Pro, Llama-3, Qwen). Building on pilot observations, future work will include:

- **Conflict-Context Stress Tests** designed to quantify Thematic Confounding,
- **Hierarchy-Reasoning Tests** to evaluate subclass–superclass identity inference,
- **Identity-Swap Bias Tests** to detect stereotyping in Bangla contexts,
- **Prompt Robustness Tests** to measure stability across zero-shot, few-shot, and CoT setups.

The pilot study demonstrated that even high-performing models can conflate event negativity with negative framing of an entity. Systematically measuring this behavior is now a primary goal.

6.0.4 Prototype Development for Real-Time Architecture (Objective 4 — Time Permitting)

If time allows, the project will begin early-stage development of a computational pipeline capable of:

- real-time ingestion of Bangla news from diverse publishers,
- multi-identity framing detection using the final model ensemble,
- automated visualization of narrative shifts and publisher-level contrasts,
- integrating HCI principles to ensure interpretability and user transparency.

Given the current research timeline, this objective will be attempted only after Objectives 1–3 are fully completed and validated.

6.0.5 Cross-Lingual and Cross-Event Generalization

Following the completion of the Bangla benchmark, a natural extension is applying the NarrativeSense framework to:

- related South Asian languages sharing cultural and political framing patterns,
- cross-event comparison (e.g., elections, communal flashpoints, student movements),
- longitudinal studies to detect shifts in media framing strategies over time.

6.0.6 Integration of Cultural Knowledge and Symbolic Context

Pilot failures confirm that LLMs struggle with implicit cues, symbolic references, and culturally encoded identity markers. Future work will investigate:

- knowledge-grounding approaches using cultural lexicons,
- event-specific metadata embeddings,
- contrastive representations that distinguish "negative event" from "negative framing".

6.0.7 Toward Responsible and Context-Aware Tools

Finally, future work includes developing responsible deployment guidelines, ensuring that:

- outputs avoid reinforcing harmful stereotypes,
- framing detections remain explainable and verifiable,
- socially sensitive identity groups are handled with caution.

6.0.8 Summary

The pilot study has clarified the path forward: expand the dataset, refine the taxonomy, strengthen annotation reliability, and construct rigorous evaluation protocols to address the failure modes uncovered by GPT-5 and Gemini 3 Pro. These steps will collectively enable the completion of the full NarrativeSense framework, with real-time computational architecture pursued if time permits.

Chapter 7

Conclusion

This thesis set out to address a critical methodological and conceptual gap in Bangla computational media analysis: the absence of a culturally grounded, theoretically informed framework for detecting identity framing in news narratives. Existing work in Bangla media analysis has focused primarily on media bias, sentiment polarity, or political leaning, without examining how social, political, religious, or institutional identities are constructed, praised, vilified, or marginalized within news discourse. The **NarrativeSense** project was designed to fill this void through a multi-stage research program involving taxonomy construction, dataset development, human annotation, and LLM evaluation.

At the current stage of the research, the project is in the middle of data collection and annotation development. However, the exploratory pilot studies conducted on a small set of news articles already provide strong evidence that identity framing—especially in conflict-heavy Bangla news contexts—is a non-trivial problem for modern Large Language Models (LLMs). These early findings validate the core motivation for NarrativeSense and offer actionable insights for completing the remaining objectives.

7.1 Summary of Achievements

Despite being in-progress, this thesis has already made several foundational contributions:

- **A comprehensive Bangla identity-framing taxonomy** grounded in cultural, political, and linguistic realities.

- **A structured multi-level annotation protocol** capturing explicit, implicit, and omission-based framing.
- **A hybrid corpus design** combining identity-guided and event-centered clustering to ensure contextual richness.
- **Pilot evaluation of LLMs** (GPT-5 and Gemini 3 Pro) that revealed crucial failure modes such as:
  - *Thematic Confounding* — models misclassifying victims as negatively framed due to conflict vocabulary.
  - *Superclass Identification Failure* — inability to map subclasses (e.g., quota activists) to super-identities (students).

These findings confirm that existing LLMs, even frontier models, struggle with the nuance, cultural signals, and implicit patterns characteristic of Bangla identity framing.

7.2 Closing Remarks

This thesis establishes the foundation for the first culturally grounded, theoretically informed identity framing benchmark for Bangla. The preliminary findings demonstrate that even state-of-the-art LLMs struggle with narrative reasoning in low-resource, culturally dense settings like Bangla news. By combining human-centered annotation, identity-event corpus design, and rigorous stress testing, **NarrativeSense** sets a new direction for responsible and context-aware computational media analysis in South Asia.

The work completed so far—supported by the roadmap outlined for future stages—positions NarrativeSense as a significant contribution to both computational linguistics and Bangla media studies, paving the way for deeper understanding of identity narratives in digital news ecosystems.

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Appendices

Appendix A

Taxonomy of Identity Classification

This appendix presents the complete taxonomy of identities used for annotation in the NarrativeSense framework. The categories reflect religious, political-ideological, demographic, institutional, issue-specific, geopolitical, and socio-professional identities, alongside their data-enriched computational tracking keywords.

Category	Target Identity	Example Keywords
Religious	Muslim	মুসলিম, ইসলাম, মুসলমান, ইমাম, আলেম, হুজুর, মাদ্রাসা, মাদরাসা, মাদরাসায়, মসজিদ, কওমি
	Hindu	হিন্দু, পূজা, সংখ্যালঘু হিন্দু, মণ্ডপ, সংখ্যালঘু, সংখ্যালঘুদের
	Buddhist & Christian	বৌদ্ধ, বৌদ্ধ মন্দির, বৌদ্ধ ভিক্ষু, খ্রিস্টান, গির্জা, খ্রিস্টান ধর্মাবলম্বী, খ্রিস্টান সম্প্রদায়
	Ahmadiyya	আহমদিয়া, কাদিয়ানী
	Baul	বাউল, মানবধর্ম, লালন, আবুল সরকার
Political & Ideological	Awami League	আওয়ামী, লীগ, ক্ষমতাসীন, ছাত্রলীগ, আ.লীগ, আ.লীগের, লীগের, ফ্যাসিবাদ, ফ্যাসিস্ট, হাসিনা, স্বৈরাচার, পতিত সরকার
	BNP	বিএনপি, ছাত্রদল, বিএনপির, যুবদল, খালেদা জিয়া, তারেক রহমান
	Jamaat-e-Islami	জামায়াতে ইসলামি, শিবির, জামায়াতে, রাজাকার, আলবদর, আল শামস, জামায়াতের

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Table A.1 (continued)

Category	Target Identity	Example Keywords
	Islamist / Conservative	হেফাজত, হেফাজতে, হেফাজতের, ইসলামী আন্দোলন
	National Citizens' Party	জাতীয় নাগরিক পার্টি, এনসিপি, জাতীয় নাগরিক পার্টির, এনসিপির, NCP
Demographic	Pahari / Indigenous	পাহাড়ি, আদিবাসী, ক্ষুদ্র নৃগোষ্ঠী, চট্টগ্রামের পাহাড়, চট্টগ্রামের পাহাড়ি, চট্টগ্রামের পাহাড়িদের, চাকমা, মারমা, সাঁওতাল, পাহাড়ী
	Rohingya	রোহিঙ্গা, রোহিঙ্গাদের, রোহিঙ্গা শরণার্থী, রোহিঙ্গা ক্যাম্প, রোহিঙ্গা জনগোষ্ঠী, উখিয়া, কুতুপালং, কুতুপালং শরণার্থী ক্যাম্প
Institutional	Government	সরকার, সরকারি, মন্ত্রিপরিষদ, মন্ত্রীরা, প্রধানমন্ত্রী, মন্ত্রীর, মন্ত্রিপরিষদের
	Military	সেনা, সেনাবাহিনী, সেনাবাহিনীর, বিমান বাহিনী, মেজর, সামরিক বাহিনী
	Law Enforcement	পুলিশ, র‍্যাব, আইনশৃঙ্খলা বাহিনী, আটক, গুম, পুলিশের, থানা, ডিবি, সিআইডি, সিটিটিসি, CTTC
	Judiciary	আদালত, আদালতে, আদালতের, বিচারপতি, বিচারক, দুদক, দুদকের
Issue-Specific & Narrative Frames	Quota Reformists	কোটা, আন্দোলনকারী, বৈষম্যবিরোধী, ছাত্র-জনতার, বৈষম্যবিরোধী ছাত্র আন্দোলনকারী, জুলাই যোদ্ধা, ইনকিলাব, পুসাব, আন্দোলনে, আন্দোলনের, সমন্বয়ক
	Islamist Framing	ইসলামপন্থী, ইসলামপন্থি, কট্টর, রক্ষণশীল, জঙ্গি, উগ্রপন্থী, তৌহিদি জনতা, সন্ত্রাসী, সন্ত্রাসবাদী, সন্ত্রাসাসের, সন্ত্রাসীদের, মৌলবাদী, মৌলবাদী জনতা, মৌলবাদী আন্দোলনকারী, মব, ওয়ার, টেরর
	Secular Framing	প্রগতিশীল, মুক্তচিন্তা, মুক্তমনা, নাস্তিক, ধর্মনিরপেক্ষ, ধর্মনিরপেক্ষতাবাদী, সেক্যুলার, সেক্যুলারিজম
	Leftist Movements	বাম, বামপন্থী, তৃণমূল বাম, তৃণমূল বামপন্থী, তৃণমূল বামপন্থীদের, ছাত্র ইউনিয়ন, সর্বহারা, মাওবাদী, কমিউনিস্ট, সিপিবি, বাসদ

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Table A.1 (continued)

Category	Target Identity	Example Keywords
	General Protesters	বিক্ষোভকারী, প্রতিবাদকারী, বিক্ষোভকারীদের, ধর্মঘটকারী, বিক্ষোভের, বিক্ষোভে, বিক্ষোভে অংশগ্রহণকারী
	Victims	ভুক্তভোগী, ক্ষতিগ্রস্ত, নিহত, আহত, গুম, গুমের, হত্যার, গণপিটুনি, নির্যাতন, হামলা, হামলার
	Narcotics / Crime	মাদক, মাদকাসক্ত, মাদক ব্যবসায়ী, মাদক কারবারি, মাদক সেবী, ইয়াবা, ফেনসিডিল, হেরোইন, কোকেন
Geopolitical	India	ভারত, ভারতপন্থী, দিল্লি, নয়া দিল্লি, ভারতীয়, ইন্ডিয়া, পুশ ইন, বিএসএফ, সীমান্ত, সীমান্তে, চোরাকারবারি
	Western Nations	আমেরিকা, মার্কিন, যুক্তরাষ্ট্র, পশ্চিমারা, পশ্চিম, ইউরোপীয় ইউনিয়ন, ওয়াশিংটন, স্যাঙ্কশন, ভিসা নীতি
	Int. Organizations	জাতিসংঘ, UN, বিশ্ব স্বাস্থ্য সংস্থা, বিশ্ব ব্যাংক, আইএমএফ, বিশ্ব বাণিজ্য সংস্থা, বিশ্ব মানবাধিকার সংস্থা
	Israel / Zionist	ইসরাইল, ইহুদী, জায়নিস্ট, যায়নিস্ট
	Palestine	জেরুজালেম, ফিলিস্তিন, হামাস, ফাতাহ
	Pakistan	পাকিস্তান, পাকিস্তানি, ইসলামাবাদ
	China & Russia	চীন, বেজিং, রাশিয়া
Social & Professional	Students & Academia	শিক্ষার্থী, ছাত্র, ছাত্রী, বিশ্ববিদ্যালয়, কলেজ, শিক্ষার্থীদের, শিক্ষার্থীরা, মেডিকেল শিক্ষার্থীদের, ইন্টার্ন চিকিৎসক, শিক্ষক
	Workers	শ্রমিক, দিনমজুর, গার্মেন্টস, পোশাক শ্রমিক
	Vulnerable Groups	সুবিধাবঞ্চিত, বস্তিবাসী, শিশু
	Intellectuals	বুদ্ধিজীবী, সুশীল সমাজ, বিশিষ্ট নাগরিক, বিশ্লেষক
	Media & Journalists	সাংবাদিক, সাংবাদিকরা, গণমাধ্যমকর্মী, প্রেস ক্লাব, প্রতিবেদক
	Legal Professionals	আইনজীবী, আইনজীবীরা, আইনজীবী সমিতি
	Business Elites	ব্যবসায়ী, ব্যবসায়ীরা, সিন্ডিকেট, শিল্পপতি, কর্পোরেট, মালিকপক্ষ, পুঁজিবাদী